Batch File Record-Count Control

1. Aim

To write a COBOL program to compare the expected and observed record counts and display `MATCH` if they are equal; otherwise, display `MISMATCH` with the absolute difference.

2. Algorithm

1. Start the program.
2. Declare variables for expected count, observed count, and difference.
3. Read the expected record count.
4. Read the observed record count.
5. Compare both counts.
6. If both counts are equal, display `MATCH`.
7. Otherwise, calculate the absolute difference and display `MISMATCH` with the difference.
8. Stop the program.

3. Approach

The program uses `ACCEPT` to read both counts and `IF-ELSE` statements to compare them. If the counts are unequal, `COMPUTE` calculates the absolute difference by subtracting the smaller count from the larger count. Finally, `DISPLAY` prints the result.

4. Program

```
IDENTIFICATION DIVISION.
PROGRAM-ID. RECORD-CHECK.

DATA DIVISION.
WORKING-STORAGE SECTION.
01 EXPECTED-COUNT PIC 9(9).
01 OBSERVED-COUNT PIC 9(9).
01 COUNT-DIFF     PIC 9(9).

PROCEDURE DIVISION.
    ACCEPT EXPECTED-COUNT.
```

```
ACCEPT OBSERVED-COUNT.

IF EXPECTED-COUNT = OBSERVED-COUNT
    DISPLAY "MATCH"
ELSE
    IF EXPECTED-COUNT > OBSERVED-COUNT
        COMPUTE COUNT-DIFF =
            EXPECTED-COUNT - OBSERVED-COUNT
    ELSE
        COMPUTE COUNT-DIFF =
            OBSERVED-COUNT - EXPECTED-COUNT
    END-IF

    DISPLAY "MISMATCH: " COUNT-DIFF
END-IF.

STOP RUN.
```

5. Output

Sample Input

```
12500
12498
```

Sample Output

```
MISMATCH: 2
```

6. Result

Thus, the COBOL program was successfully written and executed to compare the expected and observed record counts and display `MATCH` or `MISMATCH` with the absolute difference.